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Confidence-calibrated semantic resource allocation for multi-UAV low-altitude networks

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Abstract

Multi-unmanned aerial vehicle (Multi-UAV) low-altitude networks require efficient transmission of perception information under limited bandwidth and dynamic channels. Semantic communication is a promising solution. However, existing designs rarely address two critical challenges: Detector confidence is often miscalibrated, and multiple UAVs may transmit redundant semantics from overlapping views. To address these issues, we propose a confidence-calibrated visual semantic communication framework for multi-UAV networks. Each UAV extracts semantic tokens and local detection hypotheses from onboard images. It then estimates calibrated confidence and localization quality for candidate objects. Based on these cues, we define a reliable semantic value. This value integrates calibrated confidence, localization quality, class importance, and cross-UAV semantic novelty. We further formulate a joint wireless-semantic optimization problem covering subchannel assignment, power allocation, semantic token selection, and semantic quantization. To evaluate resource utilization, we introduce confidence-calibrated semantic efficiency (CCSE), which measures reliable semantic gain per unit communication cost. We solve the mixed discrete-continuous stochastic problem by using a hierarchical reinforcement learning-based resource allocation scheme. In this scheme, the upper layer allocates wireless resources, and the lower layer determines semantic content and bit depth. Extended simulation results show that the proposed framework enables efficient and trustworthy multi-UAV visual semantic communications in low-altitude networks and outperforms several baselines.

Keywords: Multi-UAV networks, Semantic communications, Low-altitude perception, Confidence calibration, Reinforcement learning, Semantic resource allocation

1 Introduction

Low-altitude intelligent networks are becoming a fundamental platform for unmanned aerial vehicle (UAV)-enabled remote sensing, emergency response, and urban supervision. In such scenarios, multiple UAVs often collaborate to observe a target area and upload perception results to an edge server for global situational awareness [1]. Compared with single-UAV operation, multi-UAV cooperation improves spatial coverage and observation diversity, thereby enhancing perception robustness [2]. However, this benefit is achieved at the cost of substantially increased uplink traffic. When multiple UAVs simultaneously transmit raw images or high-dimensional intermediate features,

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the communication overhead can rapidly exceed the available wireless capacity in bandwidth-constrained and time-varying low-altitude environments [3].

This limitation motivates the adoption of semantic communication. Rather than preserving source-level fidelity, semantic communication transmits task-relevant information for downstream inference [4]. For visual perception tasks, the receiver is concerned primarily with semantic content, such as object category, location, and contextual cues, rather than exact pixel reconstruction [5]. Consequently, semantic communication provides a suitable framework for reducing transmission overhead while retaining information that is most relevant to decision making. However, extending visual semantic communications for multi-UAV low-altitude networks remains two challenges.

The first challenge lies in the reliability of local perception outputs. Aerial observations are frequently affected by small objects, scale variation, motion blur, viewpoint distortion, and complex backgrounds [6, 7], which can make detector confidence poorly aligned with actual prediction correctness. If communication scheduling directly depends on such raw confidence, the system may prioritize unreliable observations while suppressing semantically useful ones. Therefore, confidence must be calibrated before it can serve as a meaningful basis for resource allocation [8].

The second challenge arises from semantic redundancy across UAVs [9]. Since multiple UAVs may observe overlapping regions from nearby viewpoints, the generated semantic content is often correlated. In this setting, channel quality or local confidence alone is insufficient for communication control. Even a highly confident observation may have limited utility if similar semantics have already been delivered by other UAVs [10]. Effective multi-UAV semantic communication, therefore, requires not only reliability-aware prioritization but also an explicit characterization of the incremental contribution of each UAV to the global perception task [11].

To solve the above two challenges, in this paper, the resource allocation problem for multi-UAV visual semantic communications is guided by semantic contribution rather than raw detector confidence. To this end, we propose a confidence-calibrated resource allocation framework for multi-UAV visual semantic communications. Specifically, each UAV extracts visual semantic tokens and local detection hypotheses from onboard images and estimates calibrated confidence together with localization quality for candidate objects. These factors are combined with class importance and semantic novelty to define a reliable semantic value for communication-aware prioritization. Based on this representation, the edge server performs joint wireless-semantic scheduling under a single-shot transmission regime without retransmission. To evaluate the effectiveness of resource utilization, we further introduce a new metric, termed confidence-calibrated semantic efficiency (CCSE), which measures a reliable semantic gain per unit of communication cost. On this basis, the long-term scheduling problem is formulated as a Markov decision process, and a hierarchical reinforcement learning framework is developed to optimize the coupled wireless-semantic decision process. The main contributions of this paper are summarized as follows.

- We propose a multi-UAV visual semantic communication framework for low-altitude perception, in which confidence calibration is explicitly integrated into semantic extraction and communication control.

- We define a reliable semantic value that jointly incorporates calibrated confidence, localization quality, class importance, and cross-UAV semantic novelty, thereby linking perception reliability to communication-aware semantic utility. We introduce a novel CCSE metric to evaluate the trustworthy semantic gains over limited wireless resources.
- We formulate a joint optimization problem over subchannel assignment, power control, semantic token selection, and quantization depth and develop a hierarchical reinforcement learning scheme to solve the resulting mixed discrete-continuous stochastic control problem.
- Simulation results demonstrate that the proposed framework outperforms the baseline schemes in efficient and reliable multi-UAV visual semantic communications.

2 Related work

2.1 Semantic communication for visual intelligence

Deep joint source-channel coding has provided the technical foundation for visual semantic communications. The early DeepJSCC framework in [4] demonstrated that end-to-end learned image transmission can outperform separation-based baselines in practical wireless settings, while its feedback-aware and bandwidth-agile extensions in [12, 13] further improved adaptability to channel variation and progressive transmission. Attention-guided and OFDM-oriented variants in [14] showed that neural transceivers can be made more robust to SNR mismatch, multipath fading, and non-ideal physical-layer effects. The retrieval-oriented design in [15] replaced pixel recovery with downstream matching accuracy, and the nonlinear transform-based framework in [16] incorporated learned latent priors to improve content-aware transmission. Multi-task and semantics-aware formulations in [17, 18] further considered image recovery, classification, and perceptual quality within a unified JSCC pipeline, while the explicit semantic base design in [19] introduced structured semantic carriers for image transmission. These studies collectively establish the feasibility of visual semantic transmission, but they are mainly developed for single-link or single-user scenarios and do not directly address cooperative low-altitude perception with multiple UAVs.

2.2 Communication-efficient cooperative perception

Communication-efficient cooperative perception has been extensively studied in multi-agent sensing systems. Early works such as Who2com, When2com, and V2VNet [20–22] introduced learnable communication policies for deciding who should communicate, when collaboration should be triggered, and how compressed intermediate features should be exchanged across agents. EMP [23] further incorporated edge assistance into multi-vehicle perception, showing that system-level communication scheduling can substantially improve collaborative sensing performance. Subsequent research focused more explicitly on the perception-communication trade-off. DiscoNet and Where2comm [10, 24] learned adaptive collaboration graphs and spatial confidence maps to select informative regions for transmission, while V2X-ViT and CoBEVT [25, 26] employed transformer-based fusion backbones to improve cross-agent feature aggregation. More recent studies, including HM-ViT, How2comm, and UMC [11, 27, 28], addressed hetero-modal

cooperation, pragmatic bandwidth-aware collaboration, and multi-resolution communication, respectively. These methods provide important insights into message selection and feature fusion under limited bandwidth, but they generally assume that transmitted information is already reliable and rarely model the calibration quality of local perception outputs as a communication control factor.

2.3 Confidence calibration and uncertainty-aware object detection

Reliable confidence estimation has been widely studied in uncertainty-aware object detection. BayesOD, Gaussian YOLOv3, and the uncertainty-aware detector in [29–31] modeled predictive uncertainty directly in the detection process, with particular emphasis on localization uncertainty and reliability under autonomous-driving conditions. These works demonstrated that uncertainty can be estimated together with object detections, but they primarily focused on perception reliability rather than communication-aware decision making. More recent studies have shifted from uncertainty estimation to explicit calibration analysis and calibration-oriented detector design. Conformal risk control and confidence calibration for detection were investigated in [32, 33], while train-time calibration losses and self-aware detector designs were proposed in [34, 35]. The works in [36–38] further revisited calibration metrics for detection outputs, introduced transformer-specific calibrated detectors, and analyzed pitfalls in detector calibration evaluation. Taken together, these studies show that detector confidence should reflect not only classification correctness but also localization quality; however, they still treat calibration primarily as a perception-side issue and do not integrate it into semantic communication or wireless resource allocation.

2.4 Reinforcement learning for wireless and semantic resource allocation

The authors in [39–42] utilized RL to enhance the communication and spectrum efficiency in traditional communication systems, while the semantic features of source signals are not considered. Semantic resource allocation frameworks in [43–45] considered QoE-aware semantic symbol scheduling, joint computing-and-bandwidth allocation, and adaptive semantic-bit resource control under dynamic wireless conditions. In addition, the works [46] and [47] proposed RL-based resource allocation for cross-layer semantic security.

In contrast to the above studies, this work jointly addresses confidence calibration, semantic utility modeling, wireless-semantic coupling, cross-UAV redundancy, and hierarchical reinforcement learning in a unified framework for multi-UAV visual semantic communications.

3 Semantic communication system

3.1 Communication system and semantic task model

We consider a low-altitude multi-UAV visual sensing network consisting of a ground edge server and a set of UAVs indexed by $\mathcal{U} = \{1, 2, \ldots, U\}$, as shown in Fig. 1. The UAVs collaboratively monitor a common area of interest and periodically report perception-related information to the edge server for global scene understanding. Such a system is representative of low-altitude applications including urban surveillance, infrastructure inspection, emergency reconnaissance, and traffic observation, where

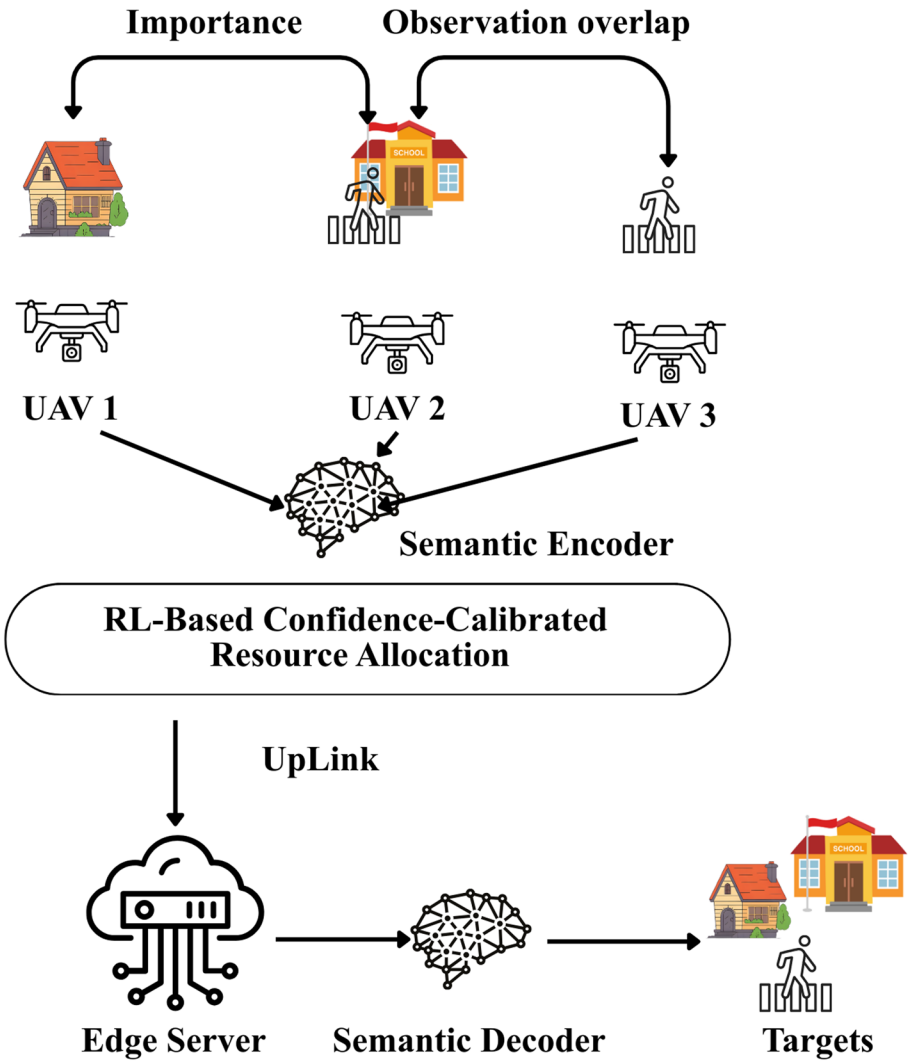


Fig. 1 The considered multi-UAV-enabled semantic communication system

the objective is to infer an accurate global perception result from multiple spatially distributed aerial viewpoints.

Time is divided into equal-duration slots indexed by $t \in \{1, 2, \ldots\}$. At slot t, each UAV $u \in \mathcal{U}$ acquires an aerial image

$$x_{u,t} \in \mathbb{R}^{H \times W \times C}, \tag{1}$$

where H, W, and C denote the image height, width, and channel dimension, respectively. Due to differences in flight position, altitude, viewing angle, and local scene content, the observations collected by different UAVs are generally heterogeneous and partially overlapping. As a result, the information captured across UAVs is complementary in some regions and redundant in others.

Based on the multi-UAV observations, the edge server aims to infer a global object set

$$\hat{\mathcal{O}}_t = \left\{ (\hat{b}_{t,k}, \hat{c}_{t,k}, \hat{s}_{t,k}) \right\}_{k=1}^{\hat{K}_t}, \tag{2}$$

where $\hat{K}_t$ is the number of detected objects at slot t, $\hat{b}_{t,k}$ denotes the bounding box of the k-th detected object, $\hat{c}_{t,k}$ denotes its semantic category, and $\hat{s}_{t,k} \in [0,1]$ denotes the corresponding fused confidence score. The goal of the system is not to reconstruct the original images collected by all UAVs, but to deliver sufficient visual semantics to support accurate global detection at the edge server under constrained uplink communication resources.

To reduce communication overhead, each UAV does not transmit raw images. Instead, it first applies a semantic encoder to extract a compact set of task-relevant semantic tokens:

$$Z_{u,t} = E_\theta(x_{u,t}) = \left\{ z_{u,t}^{(m)} \right\}_{m=1}^{N_{u,t}}, \tag{3}$$

where $E_\theta(\cdot)$ denotes the semantic encoder, $N_{u,t}$ is the number of semantic candidates generated from UAV u at slot t, and each token $z_{u,t}^{(m)} \in \mathbb{R}^d$ is a d-dimensional representation carrying compact visual semantics relevant to downstream object detection. These tokens may encode object appearance, spatial configuration, contextual information, and other task-oriented cues extracted from the aerial image.

Since the semantic content generated by different UAVs is not equally informative for global inference, the transmitted token subsets are selected under communication constraints and delivered to the edge server through uplink wireless links. Let $\hat{Z}_{u,t}$ denote the set of semantic tokens from UAV u that are successfully recovered at the receiver side after transmission and decoding. The edge server then performs multi-UAV semantic fusion according to

$$\hat{Z}_t^{\text{fuse}} = F_\psi \left(\hat{Z}_{1,t}, \hat{Z}_{2,t}, \ldots, \hat{Z}_{U,t} \right), \tag{4}$$

where $F_\psi(\cdot)$ denotes the cross-UAV semantic fusion module. This module aggregates complementary semantics from different viewpoints while suppressing redundant information, thereby constructing a global semantic representation for the monitored region.

Based on the fused semantic representation, the final global detections are obtained as

$$\hat{\mathcal{O}}_t = H_\omega \left(\hat{Z}_t^{\text{fuse}} \right), \tag{5}$$

where $H_\omega(\cdot)$ denotes the global detection head deployed at the edge server. In this work, the edge server serves as the centralized decision making unit that collects semantic information from all UAVs, performs semantic fusion, and outputs the final perception result. Consequently, the overall task can be viewed as a collaborative semantic inference problem, in which multiple UAVs provide distributed visual observations and the edge server integrates their transmitted semantics to accomplish reliable global object detection.

3.2 Confidence-calibrated local semantics

At slot t, each UAV performs local visual perception on its onboard observation and generates a set of candidate object hypotheses, denoted by

$$\mathcal{D}_{u,t} = \left\{ (b_{u,k,t}, c_{u,k,t}, p_{u,k,t}, q_{u,k,t}) \right\}_{k=1}^{K_{u,t}}, \tag{6}$$

where $K_{u,t}$ is the number of object candidates detected by UAV u at slot t. For the k-th candidate, $b_{u,k,t}$ denotes the predicted bounding box, $c_{u,k,t}$ denotes the predicted semantic category, $p_{u,k,t} \in [0,1]$ denotes the raw classification confidence, and $q_{u,k,t} \in [0,1]$ denotes the localization quality score, which reflects the geometric reliability of the predicted box and can be interpreted as a predicted intersection-over-union score or an equivalent box-quality indicator.

Although these local detection outputs provide an initial description of the observed scene, their confidence scores cannot be directly regarded as reliable measures of semantic importance. In low-altitude aerial perception, small objects, viewpoint distortion, scale variation, motion blur, occlusion, and background clutter frequently lead to overconfident or underconfident predictions. Consequently, if raw confidence is used directly for communication control, the resulting scheduler may assign excessive transmission resources to visually ambiguous observations while suppressing candidates that are more valuable to the global task. This motivates the introduction of an explicit confidence calibration mechanism before local perception outputs are used for semantic prioritization. To this end, we introduce a confidence calibration module that maps the raw detector score to a calibrated confidence value according to

$$\tilde{p}_{u,k,t} = \mathrm{Cal}\left(\mathrm{p_{u,k,t}}, \sigma_{\mathrm{u,k,t}}\right), \tag{7}$$

where $\sigma_{u,k,t}$ denotes an uncertainty descriptor produced by a lightweight uncertainty estimation head, and $\mathrm{Cal}(\cdot)$ denotes a confidence calibration function. To make the calibration module concrete and reproducible, we instantiate $\mathrm{Cal}(\cdot)$ as an uncertainty-aware temperature-scaling function. Let

$$\ell_{u,k,t} = \log \frac{p_{u,k,t}}{1 - p_{u,k,t}} \tag{8}$$

denote the raw confidence logit of the k-th candidate from UAV u at slot t. We attach a lightweight uncertainty head $G_\phi(\cdot)$ to the penultimate detection feature $f_{u,k,t} \in \mathbb{R}^{128}$ and obtain a scalar uncertainty descriptor as

$$\sigma_{u,k,t} = \mathrm{softplus}\left(\mathrm{W}_2\, \mathrm{ReLU}(\mathrm{W}_1 \mathrm{f_{u,k,t}} + \mathrm{b}_1) + \mathrm{b}_2\right), \tag{9}$$

where the uncertainty head is implemented as a two-layer MLP with dimensions $128 \to 64 \to 1$, followed by ReLU and Softplus activations. The calibrated confidence is then computed as

$$\tilde{p}_{u,k,t} = \mathrm{Sigmoid}\left(\frac{\ell_{\mathrm{u,k,t}} - \lambda \sigma_{\mathrm{u,k,t}}}{\tau}\right), \tag{10}$$

where $\tau > 0$ is a learnable temperature parameter, $\lambda \geq 0$ is a learnable uncertainty-weight parameter, and $\mathrm{Sigmoid}(\cdot)$ denotes the logistic sigmoid function. In this form,

larger predicted uncertainty suppresses overconfident detections, while the temperature parameter corrects global score sharpness. The calibration module is trained in a post hoc manner. In particular, we first train the YOLOv8-n detector on the VisDrone2019-DET training split. After detector training converges, we freeze the detector backbone and detection head and fit only the uncertainty head and the calibration parameters (τ, λ) on a held-out validation split. For each post-NMS detection, we define a binary correctness label $y_{u,k,t}$, where $y_{u,k,t} = 1$ if the predicted class is correct and the predicted bounding box matches a ground-truth object with IoU ≥ 0.5, and $y_{u,k,t} = 0$ otherwise. The calibration parameters are optimized by minimizing the binary negative log-likelihood

$$\mathcal{L}_{\text{cal}} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log \tilde{p}_i + (1-y_i)\log(1-\tilde{p}_i)]. \tag{11}$$

In our implementation, Adam is used with a learning rate of 10^{-3}, a batch size of 512, and early stopping based on the validation NLL.

Based on the calibrated local perception output, we further construct a communication-oriented measure that quantifies the semantic utility of each object candidate. Specifically, the reliable semantic value of the k-th candidate generated by UAV u at slot t is defined as

$$v_{u,k,t} = w(c_{u,k,t})\,\tilde{p}_{u,k,t}\,q_{u,k,t}\big(1-\rho_{u,k,t}\big), \tag{12}$$

where $w(c_{u,k,t})$ is a class-dependent importance weight, $\tilde{p}_{u,k,t}$ is the calibrated confidence, $q_{u,k,t}$ is the localization quality, and $\rho_{u,k,t} \in [0,1]$ denotes the redundancy score associated with the semantic content of this candidate. The class-dependent weight enables the system to emphasize categories with higher task relevance, the calibrated confidence characterizes semantic reliability, the localization quality captures geometric precision, and the redundancy term suppresses observations whose semantic contribution is largely duplicated by other UAVs. Through this formulation, a candidate is assigned a high semantic value only when it is reliable, spatially accurate, task-relevant, and non-redundant with respect to the current multi-UAV observation set.

For communication scheduling, it is also necessary to characterize the overall semantic contribution of each UAV rather than only individual candidates. We therefore define the aggregate reliable semantic value of UAV u at slot t as

$$\Gamma_{u,t} = \sum_{k=1}^{K_{u,t}} v_{u,k,t}. \tag{13}$$

The quantity $\Gamma_{u,t}$ provides a compact summary of the total reliable semantic content available at UAV u in the current slot. In the subsequent resource allocation design, this quantity serves as an important indicator for evaluating the communication priority of different UAVs under constrained wireless resources.

3.3 Semantic packetization and adaptive quantization

Although the semantic encoder generates a set of task-relevant tokens for each UAV observation, transmitting the entire token set is generally unnecessary and often infeasible under limited uplink resources. In practice, only a subset of semantic tokens is selected for transmission according to their contribution to the downstream perception task and the available communication budget. Accordingly, for UAV u at slot t, let

$$\mathcal{I}_{u,t} \subseteq \{1, 2, \ldots, N_{u,t}\} \tag{14}$$

denote the index set of semantic tokens selected for uplink delivery, where $N_{u,t}$ is the total number of candidate semantic tokens generated from the local observation. To bridge the gap between object-level semantic valuation and token-level transmission decisions, we explicitly introduce an object-to-token association mechanism. Since the semantic encoder produces a set of tokens $Z_{u,t} = \{z_{u,t}^{(m)}\}_{m=1}^{N_{u,t}}$ from the input image, each token corresponds to a local receptive field in the image space. For each object candidate $(b_{u,k,t}, c_{u,k,t})$, we associate it with a subset of semantic tokens whose receptive fields overlap with the predicted bounding box. Formally, we define the mapping

$$\mathcal{M}(k) = \{m \mid \mathrm{IoU}(\mathrm{r}_{\mathrm{u,t}}^{(\mathrm{m})}, \mathrm{b}_{\mathrm{u,k,t}}) > \delta\}, \tag{15}$$

where $r_{u,t}^{(m)}$ denotes the receptive field of token m, and δ is a predefined overlap threshold. Based on this association, we define a token-level semantic priority score that aggregates the contributions of associated object candidates:

$$\psi_{u,t}^{(m)} = \sum_{k:m\in\mathcal{M}(k)} v_{u,k,t}, \tag{16}$$

where $v_{u,k,t}$ is the reliable semantic value defined in (12). A token that is associated with multiple reliable and informative object candidates will receive a higher priority score.

For each selected token indexed by $m \in \mathcal{I}_{u,t}$, the system further assigns an adaptive quantization bit depth in order to balance semantic fidelity and transmission cost. Specifically, the quantization level is chosen from a finite bit-depth codebook [5, 48] given by

$$\mathcal{B} = \{b^{(1)}, b^{(2)}, \ldots, b^{(L)}\}, \tag{17}$$

where each element in $\mathcal{B}$ represents an admissible quantization resolution. A larger bit depth preserves more semantic detail but incurs higher transmission overhead, whereas a smaller bit depth reduces communication cost at the price of increased semantic distortion. Therefore, adaptive bit allocation provides an additional degree of freedom for semantic-aware resource control.

Given the selected bit depth $b_{u,t}^{(m)} \in \mathcal{B}$, the corresponding semantic token $z_{u,t}^{(m)}$ is quantized as

$$\hat{z}_{u,t}^{(m)} = Q\left(z_{u,t}^{(m)}, b_{u,t}^{(m)}\right), \tag{18}$$

where $Q(\cdot)$ denotes the semantic quantization operator. The quantized representation $\hat{z}_{u,t}^{(m)}$ is then packetized and prepared for uplink transmission. In this way, semantic packetization is jointly determined by token selection and quantization resolution,

both of which directly affect the final communication load and the utility of the delivered semantics.

Based on the selected token subset and the assigned quantization levels, the semantic transmission cost incurred by UAV u at slot t is defined as

$$C_{u,t}^{\mathrm{sem}} = \sum_{m \in \mathcal{I}_{u,t}} \mathrm{Cost}\left(\mathrm{b}_{\mathrm{u,t}}^{(\mathrm{m})}\right), \tag{19}$$

where $\mathrm{Cost}(\cdot)$ denotes the communication cost associated with a given quantization level. This cost function can be interpreted as the number of bits required to represent the quantized token and may also include protocol overhead related to packet headers, token indexing, or signaling information when needed. Therefore, $C_{u,t}^{\mathrm{sem}}$ provides a compact characterization of the semantic payload generated by UAV u in the current slot.

3.4 Uplink channel model

The selected semantic packets are uploaded from the UAVs to the edge server over a reuse-enabled uplink OFDMA system with N_{sc} available subchannels, where multiple UAVs can reuse the same subchannel within a slot. Let $a_{u,n,t} \in \{0,1\}$ denote the subchannel assignment variable for UAV u on subchannel n at slot t, where $a_{u,n,t} = 1$ indicates that subchannel n is allocated to UAV u, and $a_{u,n,t} = 0$ otherwise. Let $P_{u,n,t} \geq 0$ denote the corresponding transmit power. Under this setting, the wireless resource allocation for each UAV is determined jointly by subchannel assignment and power allocation across the available uplink spectrum.

For UAV u on subchannel n at slot t, the received signal-to-interference-plus-noise ratio is given by

$$\gamma_{u,n,t} = \frac{|h_{u,n,t}|^2 P_{u,n,t}}{\sigma_w^2 + \sum_{j \neq u} a_{j,n,t} |h_{j,n,t}|^2 P_{j,n,t}}, \tag{20}$$

where $h_{u,n,t}$ denotes the complex channel coefficient between UAV u and the edge server on subchannel n during slot t, and σ_w^2 denotes the noise power. The denominator consists of the thermal noise and the co-channel interference introduced by other simultaneously scheduled UAVs on the same subchannel. Accordingly, the instantaneous transmission capability of a UAV depends not only on its own channel quality and transmit power, but also on the spectrum reuse pattern across the multi-UAV uplink.

Given the subchannel assignment and the corresponding SINR values, the achievable transmission budget of UAV u within slot t is expressed as

$$B_{u,t} = T \sum_{n=1}^{N_{\mathrm{sc}}} a_{u,n,t} W_{\mathrm{sc}} \log_2\left(1 + \gamma_{u,n,t}\right), \tag{21}$$

where T denotes the slot duration and W_{sc} denotes the bandwidth of each subchannel. The quantity $B_{u,t}$ represents the maximum number of bits that can be reliably delivered by UAV u during the current slot under the allocated wireless resources. It therefore serves as the effective uplink budget available for semantic packet transmission.

Since the semantic payload generated by UAV u at slot t has already been characterized by the communication cost $C_{u,t}^{\mathrm{sem}}$ in (19), the uplink transmission must satisfy the feasibility constraint

$$C_{u,t}^{\mathrm{sem}} \le B_{u,t}, \qquad \forall u \in \mathcal{U}. \tag{22}$$

This constraint establishes the coupling between semantic-level decisions and wireless resource allocation. In particular, the selected semantic token subset and the assigned quantization levels determine the required payload size, whereas the subchannel assignment and power allocation determine the supportable uplink bit budget.

3.5 Cross-UAV redundancy modeling

In a multi-UAV visual sensing system, different UAVs may observe overlapping spatial regions from nearby viewpoints and therefore generate partially redundant semantic descriptions of the same scene content. If such redundancy is not explicitly modeled, the system may allocate communication resources to repeated semantic information, which reduces the overall efficiency of multi-UAV cooperation. Consequently, in addition to semantic reliability, it is necessary to characterize the novelty of transmitted content with respect to semantics that have already been received at the edge server.

To this end, the edge server maintains a semantic memory bank

$$\mathcal{M}_t = \left\{ m_t^{(1)}, m_t^{(2)}, \ldots, m_t^{(M)} \right\}, \tag{23}$$

where each prototype $m_t^{(j)}$ summarizes a representative semantic pattern extracted from recently received transmissions. The memory bank serves as a compact summary of the current global semantic context and provides a reference for evaluating whether newly generated semantic tokens contain genuinely new information or merely repeat content that has already been delivered.

For a candidate semantic token $z_{u,t}^{(m)}$, its redundancy score is defined by the maximum cosine similarity with respect to the memory bank:

$$\rho_{u,t}^{(m)} = \max_{j \in \{1,\ldots,M\}} \frac{\left\langle z_{u,t}^{(m)}, m_t^{(j)} \right\rangle}{\|z_{u,t}^{(m)}\|_2 \|m_t^{(j)}\|_2}. \tag{24}$$

A larger value of $\rho_{u,t}^{(m)}$ indicates that the current token is more similar to semantic content already stored in the memory bank and therefore contributes less new information to the global perception task. In contrast, a smaller redundancy score implies that the token carries relatively novel semantic information and is potentially more valuable for transmission under limited communication resources.

Memory initialization and update. We use semantic embedding similarity rather than 2D bounding-box IoU as the primary redundancy criterion for the following reasons. First, the redundancy score in this work is designed to measure semantic novelty with respect to the global semantic memory, rather than geometric overlap in a single image plane. Second, object boxes predicted by different UAVs are usually defined in their own camera views. Due to different altitudes, viewing angles, camera poses, and scale variations, 2D bounding-box IoU is not directly comparable across UAVs unless

accurate cross-view geometric registration is available. Using 2D IoU without reliable registration may incorrectly penalize useful observations from different viewpoints. Third, localization reliability has already been considered through the localization quality term $q_{u,k,t}$ in the reliable semantic value. Therefore, introducing 2D box overlap as the primary redundancy measure may partially duplicate the role of localization modeling and over-penalize semantically novel observations. For these reasons, we adopt embedding-level similarity with the semantic memory bank as the primary redundancy criterion. To make the semantic memory bank operational, we explicitly define its initialization and update mechanisms. At the beginning of each episode, the memory bank $\mathcal{M}_t$ is initialized using semantic tokens collected from the first few time slots. Specifically, we perform K-means clustering over the initial token set and use the cluster centroids as the initial prototypes.

During operation, the memory bank is updated online using newly received semantic tokens. For each incoming token $\hat{z}_{u,t}^{(m)}$, we compute its cosine similarity with all existing prototypes. If the maximum similarity exceeds a predefined threshold θ_{match}, the token is associated with the matched prototype $m_t^{(j)}$, and the prototype is updated using exponential moving averaging:

$$m_t^{(j)} \leftarrow (1-\eta)m_t^{(j)} + \eta \hat{z}_{u,t}^{(m)}, \tag{25}$$

where $\eta \in (0,1]$ is the update rate. Otherwise, the token is treated as a novel semantic pattern and is inserted into the memory bank as a new prototype.

4 Problem formulation

4.1 Decision variables

Based on the system model described above, the edge server acts as the centralized controller and determines the transmission strategy of the multi-UAV system at each slot t. Since semantic delivery is constrained jointly by wireless resources and semantic payload construction, the control variables naturally consist of two coupled components.

The first component corresponds to wireless resource allocation. Let

$$A_t = \{a_{u,n,t}\}, \qquad P_t = \{P_{u,n,t}\}, \tag{26}$$

denote the subchannel assignment matrix and the power allocation matrix at slot t, respectively. The variable $a_{u,n,t} \in \{0,1\}$ specifies whether subchannel n is allocated to UAV u, while $P_{u,n,t} \geq 0$ denotes the transmit power assigned to UAV u on subchannel n. Together, A_t and P_t determine the instantaneous uplink transmission capability of each UAV through the achievable bit budget defined in (21).

The second component corresponds to semantic packet construction. Let

$$\mathcal{I}_t = \{\mathcal{I}_{u,t}\}_{u\in\mathcal{U}}, \qquad B_t^q = \{b_{u,t}^{(m)}\}_{u,m}, \tag{27}$$

denote the token selection decision and the quantization decision at slot t, respectively. Here, $\mathcal{I}_{u,t}$ specifies the subset of semantic tokens selected by UAV u for transmission, and $b_{u,t}^{(m)} \in \mathcal{B}$ specifies the quantization bit depth assigned to the selected token $z_{u,t}^{(m)}$. These semantic-level variables determine the total payload size $C_{u,t}^{\text{sem}}$ and therefore directly affect the feasibility of transmission under the available uplink budget.

For notational convenience, the overall control action at slot t is written as

$$x_t = \left(A_t, P_t, \mathcal{I}_t, B_t^q\right). \tag{28}$$

This action simultaneously specifies how wireless resources are distributed across UAVs and how semantic content is packetized for transmission.

4.2 Confidence-calibrated semantic efficiency

To evaluate how effectively the system converts limited communication resources into useful perception information, it is necessary to introduce a task-oriented performance metric beyond conventional throughput or transmission rate. To this end, we define a new metric called confidence-calibrated semantic efficiency, abbreviated as CCSE, which measures the amount of reliable semantic gain delivered per unit communication cost.

Formally, the CCSE at slot t is defined as

$$\mathrm{CCSE_t} = \frac{\sum_{\mathrm{u=1}}^{\mathrm{U}} \sum_{\mathrm{k=1}}^{\mathrm{K_{u,t}}} \chi_{\mathrm{u,k,t}}\, \nu_{\mathrm{u,k,t}}}{\epsilon + \sum_{\mathrm{u=1}}^{\mathrm{U}} \left(\alpha \mathrm{C}_{\mathrm{u,t}}^{\mathrm{sem}} + \beta \mathrm{P}_{\mathrm{u,t}}^{\mathrm{tot}} + \gamma |\mathcal{I}_{\mathrm{u,t}}|\right)}, \tag{29}$$

where $\chi_{u,k,t} \in \{0,1\}$ indicates whether the semantic content associated with the k-th object candidate of UAV u is selected and successfully delivered in slot t, $\nu_{u,k,t}$ is the reliable semantic value defined in (12), and

$$P_{u,t}^{\mathrm{tot}} = \sum_{n=1}^{N_{\mathrm{sc}}} P_{u,n,t} \tag{30}$$

denotes the total transmit power of UAV u in the current slot. The coefficients α, β, and γ are nonnegative weighting parameters used to balance semantic payload cost, transmit power consumption, and token selection overhead, respectively. The small constant $\epsilon > 0$ is introduced to ensure numerical stability.

It is worth noting that CCSE is conceptually different from conventional physical-layer spectral efficiency. Conventional spectral efficiency characterizes the bit delivery capability of the wireless link, usually measured in bits/s/Hz, and is mainly determined by the channel gain, interference, bandwidth, and power allocation. In contrast, CCSE is a semantic-layer efficiency metric that evaluates how much reliable task-relevant semantic gain is obtained per unit communication expenditure. Therefore, CCSE is not intended to replace physical-layer spectral efficiency, but to complement it from a semantic task-oriented perspective.

The numerator in (29) represents the total reliable semantic gain actually delivered by the multi-UAV system. Since $\nu_{u,k,t}$ jointly reflects calibrated confidence, localization quality, class importance, and semantic novelty, the accumulated numerator characterizes the global semantic contribution achieved after communication scheduling. The denominator captures the corresponding communication expenditure from three aspects, namely semantic payload size, power consumption, and the number of selected tokens. Consequently, $\mathrm{CCSE_t}$ provides a compact criterion for quantifying semantic resource efficiency in a multi-UAV visual communication system.

4.3 Fairness and redundancy penalties

Although maximizing semantic efficiency is desirable, a communication strategy solely driven by instantaneous utility may produce highly unbalanced resource allocation across UAVs. In particular, the scheduler may repeatedly favor a small subset of UAVs with good channels or highly informative viewpoints, while persistently suppressing other participants. Such behavior may be detrimental to long-term cooperative perception, especially when scene conditions and observation quality evolve over time. To mitigate this issue, we introduce a fairness penalty defined as

$$\mathcal{F}_t = \mathrm{Var}\left(\Gamma^{\mathrm{tx}}_{1,\mathrm{t}}, \Gamma^{\mathrm{tx}}_{2,\mathrm{t}}, \ldots, \Gamma^{\mathrm{tx}}_{\mathrm{U},\mathrm{t}}\right), \tag{31}$$

where $\Gamma^{\mathrm{tx}}_{u,t}$ denotes the actually transmitted reliable semantic gain of UAV u at slot t. A larger value of $\mathcal{F}_t$ indicates a more uneven distribution of semantic contribution across UAVs, whereas a smaller value implies that the delivered semantic gains are more balanced.

In addition to fairness, cross-UAV semantic redundancy should also be explicitly penalized. Even when local observations are reliable, transmitting semantically repetitive content consumes uplink resources without producing proportional improvement in global perception. To account for this effect, we define the redundancy penalty as

$$\mathcal{R}_t = \sum_{u=1}^{U} \sum_{k=1}^{K_{u,t}} \chi_{u,k,t} \rho_{u,k,t}, \tag{32}$$

where $\rho_{u,k,t}$ is the object-level redundancy score introduced in the previous section. This term quantifies the total amount of semantic repetition present in the delivered content at slot t. By penalizing $\mathcal{R}_t$, the scheduler is encouraged to prioritize complementary semantic information rather than repeatedly transmitting highly similar observations.

4.4 Joint optimization objective

Let U_t denote the end-task utility obtained at slot t, which evaluates the quality of the global perception result produced by the edge server. In general form, it is expressed as

$$U_t = \mathrm{Utility}\left(\hat{\mathcal{O}}_{\mathrm{t}}, \mathcal{O}_{\mathrm{t}}\right), \tag{33}$$

where $\hat{\mathcal{O}}_t$ is the inferred global object set and $\mathcal{O}_t$ is the corresponding ground-truth object set. Depending on the specific application, Utility$(\cdot)$ may represent global detection accuracy, mean average precision, recall, or another task-level performance metric.

Based on the above definitions, the per-slot joint optimization problem is formulated as

$$\max_{x_t} \ \lambda_1 \mathrm{CCSE}_{\mathrm{t}} + \lambda_2 \mathrm{U}_{\mathrm{t}} - \lambda_3 \mathcal{R}_{\mathrm{t}} - \lambda_4 \mathcal{F}_{\mathrm{t}}, \tag{34}$$

where λ_1, λ_2, λ_3, and λ_4 are nonnegative coefficients that control the trade-off among semantic efficiency, end-task perception quality, redundancy suppression, and fairness preservation.

The optimization is subject to the following constraints:

$$\sum_{n=1}^{N_{\mathrm{sc}}} P_{u,n,t} \le P_u^{\max}, \qquad \forall u \in \mathcal{U}, \tag{35}$$

$$C_{u,t}^{\mathrm{sem}} \le B_{u,t}(A_t, P_t), \qquad \forall u \in \mathcal{U}, \tag{36}$$

$$a_{u,n,t} \in \{0,1\}, \quad P_{u,n,t} \ge 0, \qquad \forall u \in \mathcal{U}, \ \forall n, \tag{37}$$

$$b_{u,t}^{(m)} \in \mathcal{B}, \quad \mathcal{I}_{u,t} \subseteq \{1, \ldots, N_{u,t}\}, \qquad \forall u \in \mathcal{U}. \tag{38}$$

Constraint (35) limits the total transmit power of each UAV. Constraint (36) enforces the consistency between semantic payload size and the uplink transmission budget provided by wireless resource allocation. Constraint (37) specifies the feasible domain of subchannel assignment and transmit power. Constraint (38) restricts the semantic decision variables to the predefined quantization codebook and valid token index sets.

Since channel realizations, local scene complexity, semantic redundancy, and candidate object sets vary over time, the objective of the scheduler should be defined over a long-term horizon rather than on a slot-by-slot basis. The corresponding long-term optimization problem is written as

$$\max_{\pi} \ \mathbb{E}_{\pi}\left[\sum_{t=1}^{\infty} \delta^{t-1}(\lambda_1 \mathrm{CCSE_t} + \lambda_2 \mathrm{U_t} - \lambda_3 \mathcal{R}_{\mathrm{t}} - \lambda_4 \mathcal{F}_{\mathrm{t}})\right], \tag{39}$$

where π denotes the control policy adopted by the edge server and $\delta \in (0,1)$ is the discount factor. The expectation is taken over the stochastic evolution of the wireless environment, UAV observations, semantic candidate sets, and policy-induced actions.

Problem (39) is challenging for several reasons. First, the decision space contains both discrete variables and continuous variables, which makes the action space inherently hybrid [39]. Second, the number of candidate semantic units varies across UAVs and across time slots, leading to a non-uniform semantic decision structure. Third, wireless scheduling and semantic packetization are coupled through the transmission budget constraint in (36). Fourth, the quantities that define semantic utility, including calibrated confidence and redundancy, evolve dynamically with the observation process and previously delivered content. These characteristics make the problem difficult to solve by conventional model-based optimization or static heuristic design, thereby motivating the reinforcement learning framework developed in the next section.

5 Methods

5.1 Hierarchical Markov decision process

The long-term optimization problem in (39) is a sequential decision making problem under stochastic channel conditions, time-varying scene content, and dynamically evolving semantic redundancy. Since the control variables include both wireless resource allocation and semantic packet construction, and since the semantic candidate set varies across UAVs and time slots, direct optimization by conventional static methods is

difficult. To address this issue, we formulate the decision process as a Markov decision process with a centralized agent deployed at the edge server.

At slot t, the agent observes the current system state s_t, executes an action a_t, and receives an immediate reward r_t, and then, the environment transitions to the next state s_{t+1}. In order to maintain consistency with the optimization problem defined in Sect. 4, the MDP action a_t corresponds to the joint control decision x_t, that is, the combined wireless-semantic scheduling strategy applied at slot t.

A direct treatment of the full action space is undesirable since the decision variables are hybrid in nature and strongly coupled. Specifically, subchannel assignment and token selection are discrete, whereas transmit power is continuous; meanwhile, semantic decisions must additionally satisfy the slot-dependent bit-budget constraint induced by wireless allocation. To better reflect this structure, we adopt a two-layer hierarchical decision architecture. The upper layer determines the wireless transmission configuration, including UAV scheduling, subchannel assignment, and power allocation. Conditioned on the resulting uplink transmission budget, the lower layer determines the semantic transmission strategy, including semantic token selection and token-level quantization depth. This hierarchical decomposition follows the intrinsic dependency between communication resources and semantic payload construction and provides a structured way to reduce the complexity of policy learning.

5.2 State representation

The state representation should capture both the wireless condition of each UAV and the semantic utility of its current observation. To this end, for UAV u at slot t, we first construct a slot-level summary vector

$$g_{u,t} = \left[\bar{h}_{u,t}, \Gamma_{u,t}, \bar{\tilde{p}}_{u,t}, \bar{q}_{u,t}, \bar{\rho}_{u,t}, E_{u,t}, Q_{u,t}\right], \tag{40}$$

where $\bar{h}_{u,t}$ denotes a summary of the uplink channel condition, such as the average channel gain or the average signal-to-interference-plus-noise ratio, $\Gamma_{u,t}$ denotes the aggregate reliable semantic value defined in (13), $\bar{\tilde{p}}_{u,t}$ denotes the average calibrated confidence over current object candidates, $\bar{q}_{u,t}$ denotes the average localization quality, $\bar{\rho}_{u,t}$ denotes the average redundancy score, $E_{u,t}$ denotes the residual onboard energy, and $Q_{u,t}$ denotes the semantic queue length or, equivalently, the number of candidate semantic units awaiting possible transmission.

In addition to slot-level summaries, the decision process should also preserve finer-grained object-level information, since semantic scheduling is ultimately influenced by the properties of individual object candidates. Accordingly, for the k-th object candidate at UAV u, we define the object-level descriptor

$$d_{u,k,t} = \left[\tilde{p}_{u,k,t}, q_{u,k,t}, \rho_{u,k,t}, w(c_{u,k,t})\right]. \tag{41}$$

This descriptor characterizes the reliability, geometric quality, redundancy, and task relevance of the candidate. Because the number of object candidates $K_{u,t}$ may vary across UAVs and time slots, a fixed-dimensional representation cannot be obtained by direct concatenation. We therefore employ a permutation-invariant set encoder $S_\eta(\cdot)$ to map the candidate set to a compact embedding:

$$\bar{g}_{u,t}^{\mathrm{obj}} = S_\eta\left(\{d_{u,k,t}\}_{k=1}^{K_{u,t}}\right). \tag{42}$$

The full system state is then represented as

$$s_t = \left(\{\bar{g}_{u,t}^{\mathrm{obj}}, g_{u,t}\}_{u=1}^{U}, \mathcal{M}_t\right), \tag{43}$$

where $\mathcal{M}_t$ is the semantic memory bank introduced in the redundancy modeling section. This state representation jointly captures instantaneous channel quality, local semantic value, object-level reliability, semantic redundancy, residual energy, and the global semantic context accumulated at the edge server.

5.3 Action representation

In accordance with the hierarchical MDP structure, the joint action is factorized into a wireless action and a semantic action:

$$a_t = \left(a_t^{\mathrm{w}}, a_t^{\mathrm{s}}\right). \tag{44}$$

The wireless action determines the transmission capability available to each UAV in the current slot, whereas the semantic action determines how this capability is utilized for semantic packet delivery.

The upper-layer wireless action is defined as

$$a_t^{\mathrm{w}} = (A_t, P_t), \tag{45}$$

where $A_t = \{a_{u,n,t}\}$ and $P_t = \{P_{u,n,t}\}$ denote the subchannel assignment and power allocation, respectively. Once a_t^{w} is determined, the edge server can compute the corresponding bit budget $B_{u,t}$ for each UAV through (21). Therefore, the output of the upper layer determines the feasible communication region for the subsequent semantic decision.

The lower-layer semantic action is defined as

$$a_t^{\mathrm{s}} = (\mathcal{I}_t, B_t^q), \tag{46}$$

where $\mathcal{I}_t = \{\mathcal{I}_{u,t}\}_{u\in\mathcal{U}}$ denotes the selected semantic token subsets and $B_t^q = \{b_{u,t}^{(m)}\}_{u,m}$ denotes the token-level quantization decisions. These variables determine the semantic payload size $C_{u,t}^{\mathrm{sem}}$ and therefore must satisfy the transmission feasibility condition

$$\sum_{m\in\mathcal{I}_{u,t}} \mathrm{Cost}\left(\mathrm{b}_{\mathrm{u,t}}^{(\mathrm{m})}\right) \le \mathrm{B_{u,t}}, \qquad \forall \mathrm{u} \in \mathcal{U}. \tag{47}$$

This constraint explicitly links the lower-layer semantic action to the wireless transmission capability determined by the upper layer.

5.4 Confidence-guided semantic prioritization

The semantic action is not performed uniformly across all candidates, since different object hypotheses contribute unequally to the global perception task. To characterize their relative transmission priority, we introduce a confidence-guided semantic prioritization score for the k-th object candidate of UAV u at slot t:

$$\phi_{u,k,t} = \tilde{p}_{u,k,t}^{\alpha_1} q_{u,k,t}^{\alpha_2} \left(1 - \rho_{u,k,t}\right)^{\alpha_3} w(c_{u,k,t})^{\alpha_4}, \tag{48}$$

where $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0$ are tunable exponents.

The above formulation is aligned with the reliable semantic value defined in (12), but is introduced here as an explicit prioritization mechanism for semantic packet construction. The factor $\tilde{p}_{u,k,t}$ reflects the calibrated reliability of the candidate, $q_{u,k,t}$ describes localization quality, $1 - \rho_{u,k,t}$ characterizes semantic novelty, and $w(c_{u,k,t})$ accounts for class-dependent task importance. A larger value of $\phi_{u,k,t}$ indicates that the corresponding candidate is more reliable, more informative, and more relevant to the global task. Consequently, candidates with larger prioritization scores are more likely to be selected for transmission and to receive finer quantization resolution under the available communication budget.

5.5 Reward design

The reward function should encourage communication policies that achieve high semantic utility while maintaining efficiency, diversity, and fairness. To this end, the immediate reward at slot t is defined as

$$r_t = \lambda_1 \mathrm{CCSE_t} + \lambda_2 \mathrm{U_t} - \lambda_3 \mathcal{R}_\mathrm{t} - \lambda_4 \mathcal{F}_\mathrm{t} - \lambda_5 \sum_{\mathrm{u}=1}^{\mathrm{U}} \mathrm{P}_{\mathrm{u,t}}^{\mathrm{tot}}, \tag{49}$$

where $P_{u,t}^{\mathrm{tot}} = \sum_{n=1}^{N_{\mathrm{sc}}} P_{u,n,t}$ denotes the total transmit power of UAV u in slot t.

The first term in (49) promotes efficient conversion of communication resources into reliable semantic gains through the metric $\mathrm{CCSE_t}$. The second term preserves end-task perception quality by rewarding accurate global inference. The third term suppresses semantically repetitive transmissions through the redundancy penalty $\mathcal{R}_t$, while the fourth term discourages persistently imbalanced scheduling across UAVs through the fairness penalty $\mathcal{F}_t$. The final term controls energy expenditure and prevents excessively aggressive power allocation. By combining these objectives, the reward design provides a unified criterion for long-term wireless-semantic scheduling.

5.6 Hierarchical hybrid actor–critic algorithm

Since the action space contains both discrete variables and continuous variables, we employ a hierarchical hybrid actor–critic architecture to learn the control policy. The policy is factorized as

$$\pi(a_t|s_t) = \pi_{\theta_\mathrm{w}}(a_t^{\mathrm{W}}|s_t)\, \pi_{\theta_\mathrm{s}}(a_t^{\mathrm{S}}|s_t, a_t^{\mathrm{W}}), \tag{50}$$

where π_{θ_w} denotes the upper-layer wireless policy and π_{θ_s} denotes the lower-layer semantic policy. The upper-layer policy outputs the wireless control variables, including subchannel assignment and power allocation. Conditioned on these decisions, the lower-layer policy generates the semantic packet construction strategy under the available per-UAV transmission budgets.

To evaluate the policy, we adopt two critic networks

$$Q_{\phi_1}(s_t, a_t), \qquad Q_{\phi_2}(s_t, a_t), \tag{51}$$

and use their minimum value to mitigate overestimation:

$$Q_{\min}(s_t, a_t) = \min\left\{Q_{\phi_1}(s_t, a_t), Q_{\phi_2}(s_t, a_t)\right\}. \tag{52}$$

The soft Bellman target is then written as

$$y_t = r_t + \delta\, \mathbb{E}_{a_{t+1}\sim\pi}\left[Q_{\min}(s_{t+1}, a_{t+1}) - \alpha \log \pi(a_{t+1}|s_{t+1})\right], \tag{53}$$

where α denotes the entropy temperature parameter. Based on this target, the critic objective is defined as

$$\mathcal{L}_{\text{critic}} = \sum_{j=1}^{2} \mathbb{E}\left[\left(Q_{\phi_j}(s_t, a_t) - y_t\right)^2\right]. \tag{54}$$

The actor is updated by maximizing the entropy-regularized expected return, which is equivalently written as minimizing

$$\mathcal{L}_{\text{actor}} = \mathbb{E}_{s_t}\left[\mathbb{E}_{a_t\sim\pi}\left(\alpha \log \pi(a_t|s_t) - Q_{\min}(s_t, a_t)\right)\right]. \tag{55}$$

The overall procedure is summarized in Algorithm 1.

Algorithm 1 Hierarchical RL for Multi-UAV Semantic Resource Allocation

Initialize upper-layer actor π_{θ_w}, lower-layer actor π_{θ_s}, critics Q_{ϕ_1}, Q_{ϕ_2}, target critics, replay buffer $\mathcal{D}$, and semantic memory $\mathcal{M}$.
for each episode **do**
  Reset environment and initialize slot index $t \leftarrow 1$.
  while episode not terminated **do**
    Observe system state s_t.
    Upper actor outputs wireless action $a_t^{\text{w}} = (A_t, P_t)$.
    Compute achievable bit budgets $\{B_{u,t}\}_{u=1}^{U}$ using (21).
    Lower actor outputs semantic action $a_t^{\text{s}} = (\mathcal{I}_t, B_t^q)$ under feasible action masking.
    Execute single-shot semantic transmission without retransmission.
    Reconstruct semantic tokens, fuse received semantics, and perform global detection.
    Compute CCSE_t, U_t, $\mathcal{R}_t$, $\mathcal{F}_t$, and reward r_t using (49).
    Observe next State s_{t+1} and store transition (s_t, a_t, r_t, s_{t+1}) in replay buffer $\mathcal{D}$.
    Sample a mini-batch from $\mathcal{D}$.
    Update critics and actors by minimizing (54) and (55).
    Update semantic memory $\mathcal{M}$.
    $t \leftarrow t + 1$.
  end while
end for

5.7 Computational complexity and real-time feasibility

The proposed hierarchical reinforcement learning framework follows an offline training and online inference paradigm. The training process is conducted at the edge server using historical interaction data, while the real-time deployment only requires forward inference of the learned policies.

Inference complexity. At each time slot, the decision process consists of two stages. The upper-layer policy generates wireless resource allocation decisions, with computational complexity approximately given by

$$\mathcal{O}(U \cdot N_{\mathrm{sc}}), \tag{56}$$

where U is the number of UAVs and N_{sc} is the number of subchannels. The lower-layer policy performs semantic token selection and quantization, with complexity

$$\mathcal{O}\left(\sum_{u=1}^{U} N_{u,t}\right), \tag{57}$$

where $N_{u,t}$ is the number of candidate tokens for UAV u. Therefore, the overall inference complexity scales linearly with the system size.

Inference latency. In our implementation, the actor networks consist of lightweight multilayer perceptrons. The average inference time per slot is below 5 ms on an NVIDIA RTX 4090 GPU for the default setting with $U = 4$ UAVs and $N_{u,t} \leq 64$. This latency is significantly smaller than the typical sensing and transmission interval (tens of milliseconds), indicating that the proposed framework can support real-time decision making.

Training and convergence. The training phase is performed offline and does not affect real-time operation. We observe that the hierarchical structure accelerates convergence compared to flat reinforcement learning policies, as it decomposes the joint decision space into structured subproblems. In practice, the training converges within approximately 10^5 interaction steps, after which the policy remains stable.

Overall, the proposed HRL framework achieves a favorable trade-off between decision quality and computational overhead, making it suitable for deployment in resource-constrained multi-UAV systems.

6 Results

This section presents the simulation settings, benchmark schemes, evaluation metrics, and experimental protocol used to assess the proposed confidence-calibrated multi-UAV visual semantic communication framework.

6.1 Simulation settings

The visual perception task is constructed based on the VisDrone2019-DET dataset, which is a drone-view object detection benchmark with significant scale variation, occlusion, and viewpoint diversity. The default number of UAVs is set to $U = 4$. The monitored area is set to 1000 m $\times$ 1000 m. The UAV altitude is uniformly sampled from [80, 120] m, and the horizontal speed is sampled from [8, 16] m/s. The inter-UAV overlap ratio is controlled by adjusting the relative deployment positions, with a default average overlap ratio of 0.35. The uplink adopts an OFDM transmission structure with $N_{\mathrm{sc}} = 16$ subchannels. The bandwidth of each subchannel is set to $W_{\mathrm{sc}} = 180$ kHz, resulting in a total uplink bandwidth of 2.88 MHz. The carrier frequency is set to 3.5 GHz. The uplink channel follows a distance-dependent large-scale path-loss model combined with small-scale Rician fading. The path-loss exponent is set to 2.2, the Rician factor is set to 6 dB, and the log-normal shadowing standard deviation is set to 4 dB. The thermal noise

spectral density is fixed at -174 dBm/Hz. The maximum transmit power of each UAV is $P_u^{\max} = 23$ dBm.

For visual perception, all images are resized to 640×640. The local detector is implemented using a lightweight YOLOv8-n backbone, which is first pre-trained and then fine-tuned on VisDrone2019-DET. The detector output is used to construct local object candidates and semantic tokens. The semantic encoder produces token representations of dimension $d = 128$, and the maximum number of candidate tokens generated per UAV per slot is set to $N_{u,t}^{\max} = 64$. The semantic memory bank size is set to $M = 64$. The quantization codebook is defined as $\mathcal{B} = \{2, 4, 6, 8\}$ bits. All methods are implemented in Python with PyTorch and trained on a single NVIDIA RTX 4090 GPU.

6.2 Benchmark schemes

To evaluate the proposed method, we consider two reinforcement learning-based baselines and one traditional mathematical optimization baseline, all implemented under the same detector, semantic encoder, quantization codebook, and channel simulator. The first RL baseline, denoted as HARL-SemCom, is adapted from the hybrid-action reinforcement learning framework for UAV-assisted semantic communication and employs a single-layer hybrid policy to jointly determine subchannel assignment, transmit power, token selection, and quantization depth, but does not explicitly model calibrated confidence or cross-UAV semantic redundancy. The second RL baseline, denoted as ARA-DRL, is adapted from the adaptive resource allocation framework for semantic communication networks and jointly optimizes semantic-bit quantization and radio resources through a flat DRL policy, without using the proposed hierarchical scheduling mechanism or confidence-calibrated semantic prioritization. The traditional optimization baseline, denoted as Lyapunov-SCA, is adapted from a cross-layer semantic resource allocation framework based on Lyapunov optimization, successive convex approximation, and KKT-based updates and performs slot-wise semantic admission control and power allocation under the same bit-budget constraint while ranking semantic candidates according to raw detector confidence.

6.3 Result analysis

As shown in Table 1, the proposed method remains stable under moderate variations of the reward coefficients. The default setting achieves $mAP@50 = 44.1\%$ and $\mathrm{CCSE} = 1.48$, which is consistent with the main simulation results. Increasing λ_1 improves CCSE from 1.39 to 1.55, but slightly decreases mAP@50 because the policy becomes more efficiency-oriented. Increasing λ_2 improves mAP@50 from 42.8% to 44.9%, while CCSE slightly decreases due to the higher communication cost required for perception accuracy. A larger λ_3 effectively reduces the redundancy ratio from 0.28 to 0.19, confirming the role of the redundancy penalty. The fairness coefficient λ_4 has a relatively mild effect on mAP@50 and CCSE, indicating that fairness regulation does not significantly degrade the overall system performance.

As illustrated in Fig. 2, the proposed HH-SAC scheme consistently achieves the highest detection accuracy over the entire bandwidth range. Specifically, at the default uplink bandwidth of 2.88 MHz, the proposed method attains an mAP@50 of 44.1%, which

Table 1 Sensitivity analysis of reward weighting coefficients

Varied coefficient	Value	mAP@50 (%)	CCSE	Redundancy ratio
Default setting	–	44.1	1.48	0.23
λ_1	0.5	44.7	1.39	0.25
λ_1	1.0	44.1	1.48	0.23
λ_1	1.5	43.5	1.55	0.22
λ_2	0.5	42.8	1.52	0.24
λ_2	1.0	44.1	1.48	0.23
λ_2	1.5	44.9	1.41	0.24
λ_3	0.5	44.3	1.43	0.28
λ_3	1.0	44.1	1.48	0.23
λ_3	1.5	43.7	1.50	0.19
λ_4	0.5	44.4	1.49	0.24
λ_4	1.0	44.1	1.48	0.23
λ_4	1.5	43.8	1.46	0.23

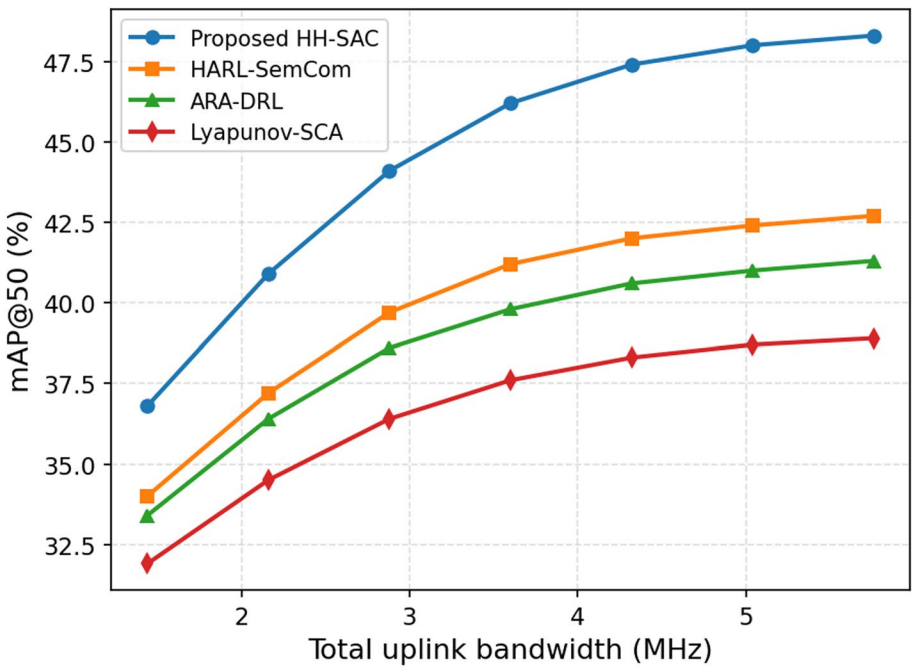


Fig. 2 mAP@50 versus total uplink bandwidth for different resource allocation schemes

improves upon HARL-SemCom, ARA-DRL, and Lyapunov-SCA by 11.1%, 14.2%, and 21.2%, respectively. When the bandwidth increases to 5.76 MHz, the mAP@50 further reaches 48.3%, still exceeding the three baselines by 13.1%, 16.9%, and 24.2%, respectively. This advantage indicates that the proposed method can utilize additional communication resources more effectively than the benchmark schemes. The main reason is that the proposed framework does not allocate resources solely according to channel quality or raw detector confidence. Instead, it jointly considers calibrated confidence, localization quality, and semantic novelty when constructing the reliable semantic value, such that the additional bandwidth is preferentially used to deliver semantically informative

and trustworthy content. As a result, the proposed scheme achieves both higher semantic fidelity and higher task utility under the same communication budget.

Figure 3 compares the confidence-calibrated semantic efficiency under different numbers of UAVs. The proposed method consistently outperforms all benchmark schemes and exhibits the most favorable scalability trend. At $U = 4$, which corresponds to the default system setting, the proposed method achieves a CCSE of 1.48, yielding gains of 14.7%, 20.3%, and 28.7% over HARL-SemCom, ARA-DRL, and Lyapunov-SCA, respectively. The CCSE further increases to 1.61 at $U = 6$, corresponding to improvements of 19.3%, 25.8%, and 36.4% over the three baselines. When the number of UAVs increases to 8, the CCSE of all methods slightly decreases, which is consistent with the fact that larger UAV populations introduce stronger semantic redundancy and more severe competition for limited wireless resources. Nevertheless, the proposed scheme remains superior because the hierarchical decision structure explicitly separates wireless allocation from semantic packet construction, thereby improving policy scalability, while the redundancy-aware semantic value suppresses repeated transmissions among overlapping UAV observations.

The effect of observation overlap is further examined in Fig. 4, where the redundancy ratio increases monotonically with the average observation overlap ratio for all methods. This trend is expected, since stronger spatial overlap among UAV observations naturally leads to more repeated semantic content. However, the proposed method maintains the lowest redundancy ratio throughout the entire range. At an overlap ratio of 0.5, the redundancy ratio of the proposed method is reduced by 23.3%, 28.3%, and 35.3% relative to HARL-SemCom, ARA-DRL, and Lyapunov-SCA, respectively. When the overlap ratio further increases to 0.7, the corresponding reductions remain significant at 21.3%, 27.3%, and 34.2%. These results confirm the necessity of explicitly modeling cross-UAV

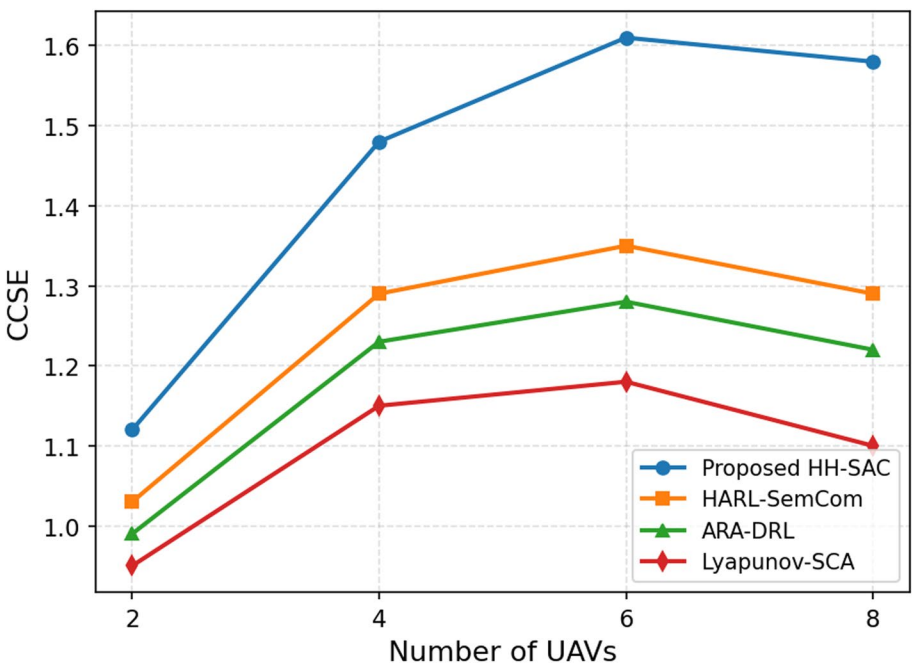


Fig. 3 Confidence-calibrated semantic efficiency versus the number of UAVs

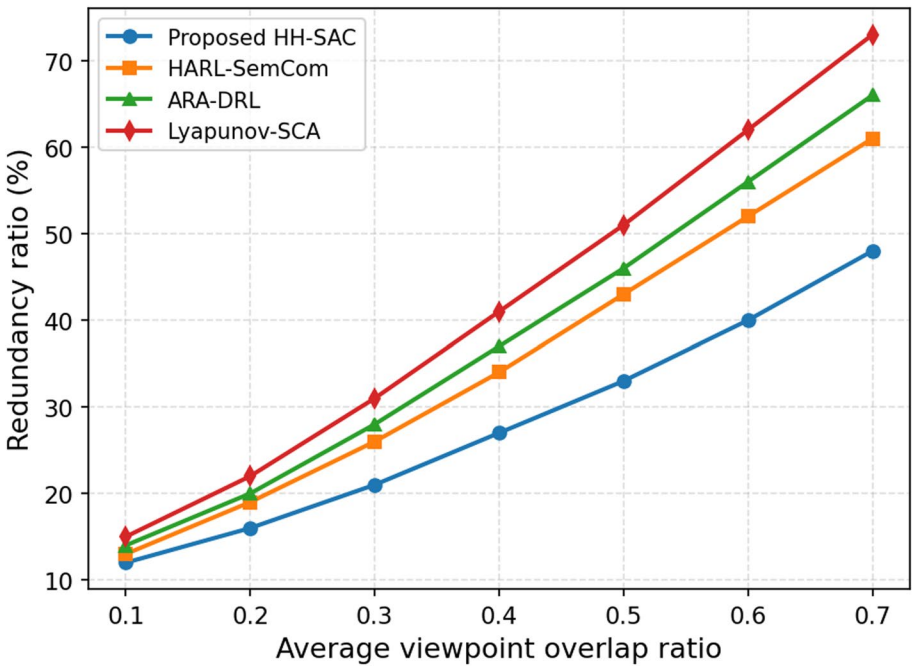


Fig. 4 Redundancy ratio versus average observation overlap ratio under different schemes

redundancy. In contrast to the benchmark schemes, the proposed method exploits the semantic memory bank and the redundancy term in the reliable semantic value, allowing the scheduler to distinguish globally informative semantics from locally reliable but repetitive observations.

The communication overhead comparison is shown in Fig. 5. The proposed method yields the lowest average uplink semantic payload, namely 112 kbit/slot, whereas HARL-SemCom, ARA-DRL, and Lyapunov-SCA require 126, 133, and 141 kbit/slot, respectively. Therefore, the proposed scheme reduces the average semantic payload by 11.1%, 15.8%, and 20.6% compared with the three baselines. Figure 5 shows that the proposed method improves performance not by transmitting more data, but by transmitting more valuable data. This behavior is a direct consequence of the confidence-calibrated semantic prioritization and the CCSE-driven reward design, which jointly encourage the policy to allocate resources to semantic units with higher expected contribution to the global perception task while avoiding transmission of redundant content.

7 Conclusion

This paper investigated confidence-calibrated resource allocation for multi-UAV visual semantic communications in low-altitude networks. To address the mismatch between local detector confidence and actual semantic utility, we introduced a reliable semantic value that jointly captures calibrated confidence, localization quality, class importance, and cross-UAV redundancy. On this basis, a joint wireless-semantic optimization problem was formulated, in which subchannel assignment, transmit power, semantic token selection, and quantization depth are optimized under a single-shot transmission constraint. To evaluate the efficiency of semantic resource utilization, we further introduced the confidence-calibrated semantic efficiency metric, which measures the

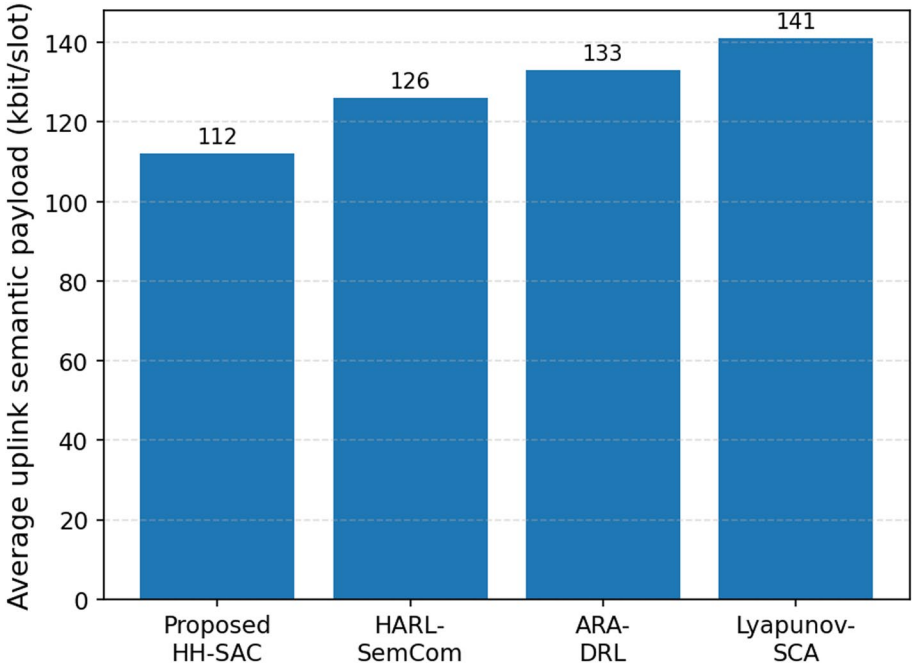


Fig. 5 Average uplink semantic payload of different schemes

reliable semantic gain achieved per unit communication cost. Given the hybrid and time-varying nature of the resulting optimization problem, a hierarchical reinforcement learning framework was developed to perform coordinated wireless-semantic scheduling. The upper layer determines wireless resource allocation, while the lower layer constructs semantic packets according to the available transmission budget and the confidence-guided prioritization mechanism. The simulation results demonstrated that the proposed method consistently achieves a better trade-off among perception accuracy, semantic efficiency, redundancy suppression, and communication overhead than the benchmark schemes. These results confirm that confidence calibration and redundancy-aware semantic modeling are both critical for effective multi-UAV semantic communication.

7.1 Discussion

The results demonstrate that effective multi-UAV semantic communication depends not only on bandwidth allocation, but also on how reliably semantic information reflects true task utility. By incorporating confidence calibration, semantic novelty, and task relevance into the transmission policy, the proposed framework improves detection performance while reducing redundant semantic traffic. In contrast to prior visual semantic communication studies that mainly focus on single-link efficiency, and cooperative perception methods that emphasize feature sharing or communication sparsification, this work shows that reliability-aware semantic valuation is critical in distributed aerial sensing. More broadly, the findings suggest that confidence calibration should be viewed not only as a perception problem, but also as an important mechanism for communication control in task-oriented wireless systems.

Several limitations should also be noted. First, the study is validated in simulation on VisDrone2019-DET and a modeled wireless uplink, so real-world deployment factors such as hardware constraints, flight dynamics, and protocol overhead are not fully captured. Second, the calibration module is treated in a general manner, and the relative effects of different calibration strategies are not separately evaluated. Third, the framework assumes centralized scheduling and focuses on object detection, which may limit direct generalization to larger, decentralized UAV swarms or to other tasks such as tracking, segmentation, and multimodal perception. Future work should therefore include real-world experiments, comparison of calibration methods, and extensions to decentralized and temporally aware semantic communication systems.

Abbreviations

UAV	Unmanned aerial vehicle
SNR	Signal-to-noise ratio
DRL	Deep reinforcement learning
SINR	Signal-to-interference-plus-noise ratio

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Author contributions
Wei Zheng performed the system modeling, writing, optimization algorithms, and implementation and participated in drafting the manuscript. Jun Li performed the optimization model design, along with optimization algorithms. Yang Zuo performed writing, optimization algorithms, and implementation. All authors have read and approved the final manuscript.

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Data availability
Data sharing is not applicable to this study.

Declarations

Competing interests
The author declares no competing interests related to this research.

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